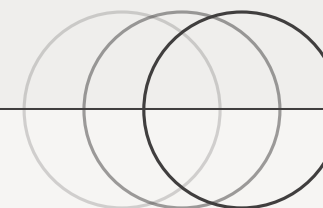


# HEALTHCARE RECOMMENDATION SYSTEM

GROUP 12

# Members



NGUYỄN VŨ ANH KHOA

NGUYỄN PHI LONG

HỒ NGUYỄN THIÊN VŨ

PHẠM HỒNG TRÀ

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# Content

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- I/ INTRODUCTION
- II/ DATASET
- III/ METHODOLOGY
- IV/ EXPERIMENTS & RESULTS
- V/ CONCLUSION

# Introduction

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## Context:

- The increasing use of LLMs for **online health-related questions**.
- **Hallucination** in LLM models.

## Project Goals:

- Developing a mechanism for **natural language understanding**
- Orienting user to **appropriate Specialist** and providing **references advices**
- **Mitigating hallucination** and quality control via **RAG**

# Introduction

## PhoBERT: Pre-trained language models for Vietnamese

Dat Quoc Nguyen<sup>1</sup> and Anh Tuan Nguyen<sup>2,\*</sup>

<sup>1</sup>VinAI Research, Vietnam; <sup>2</sup>NVIDIA, USA

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### Abstract

We present **PhoBERT** with two versions—**PhoBERT<sub>base</sub>** and **PhoBERT<sub>large</sub>**—the *first* public large-scale monolingual language models pre-trained for Vietnamese. Experimental results show that PhoBERT consistently outperforms the recent best pre-trained multilingual model XLM-R (Conneau et al., 2020) and improves the state-of-the-art in multiple Vietnamese-specific NLP tasks including Part-of-speech tagging, Dependency parsing, Named-entity recognition and Natural language inference. We release PhoBERT to facilitate future research and downstream applications for Vietnamese NLP. Our PhoBERT models are available at: <https://github.com/VinAIRResearch/PhoBERT>.

models except XLM-R. It is worth noting that Wikipedia data is not representative of a general language use, and the Vietnamese Wikipedia data is relatively small (1GB in size uncompressed), while pre-trained language models can be significantly improved by using more pre-training data (Liu et al., 2019).

- All publicly released monolingual and multilingual BERT-based language models are not aware of the difference between Vietnamese syllables and word tokens. This ambiguity comes from the fact that the white space is also used to separate syllables that constitute words when written in Vietnamese.<sup>1</sup> For example, a 6-syllable written text “Tôi là một nghiên cứu viên” (I am a researcher) forms 4 words “Tôi,

## A STUDY OF TEXT NORMALIZATION IN VIETNAMESE FOR TEXT-TO-SPEECH SYSTEM

Dinh Anh Tuan, Phi Tung Lam, Phan Dang Hung

Institute of Information Technology

### ABSTRACT

Text Normalization plays an important role in a Text-To-Speech (TTS) system. There are many researches on the topic in recent years. But we still have had many problems to deal with. In Vietnam, developing TTS system has had many positive results but researching Text Normalization hasn't got necessary attention. Many TTS system in Vietnamese can only process the input text in pronounceable forms or they use some self-proposed rules to solve particular cases. In order to increase the quality of Text Normalization, we need a more methodical approach than old ones. In this paper, we propose a classification of Vietnamese Non Standard Words (NSWs) based on our training set and we also propose a model to normalize NSWs, using powerful methods such as regular expression and we use n-gram model, decision lists. In the paper, we built a training set of 2000 Vietnamese articles of Wikipedia. The training data has about 9 millions syllables with more than 11 percent of NSWs. There is an average of 75 sentences per article. Each sentence has 20 to 30 syllables. We use regular expression and decision list for Vietnamese. Our decision lists have 1383 collocations. Our test set includes 2000 random sentences from training data and 2000 random sentences from [www.tinngan.vn](http://www.tinngan.vn) and 2000 sentences from [www.tuoiitre.vn](http://www.tuoiitre.vn). The accuracy of the experiment is more than 98.18 %.

**Index Terms**— Text Normalization, Speech Synthesis

First of all, a training set containing 2000 Vietnamese articles from Wikipedia is prepared. Then we observed the training set to figure out the possible types of NSWs. The categorization decides the range of NSWs in our works. We also learn the remarkable properties of each type. Based on our observation, we decide to use regular expression to identify the type of each NSWs and n-gram model to deal with local context problem.

In the paper, Section 2 describes a possible classification of NSWs in Vietnamese. Section 3 presents a model to normalize text. In section 4, Decision List, a homograph dissambiguating technique, is described. In Section 5, an experience is done. The section presents our training set, test set and results of the experience.

### 2. A CLASSIFICATION OF NSWs

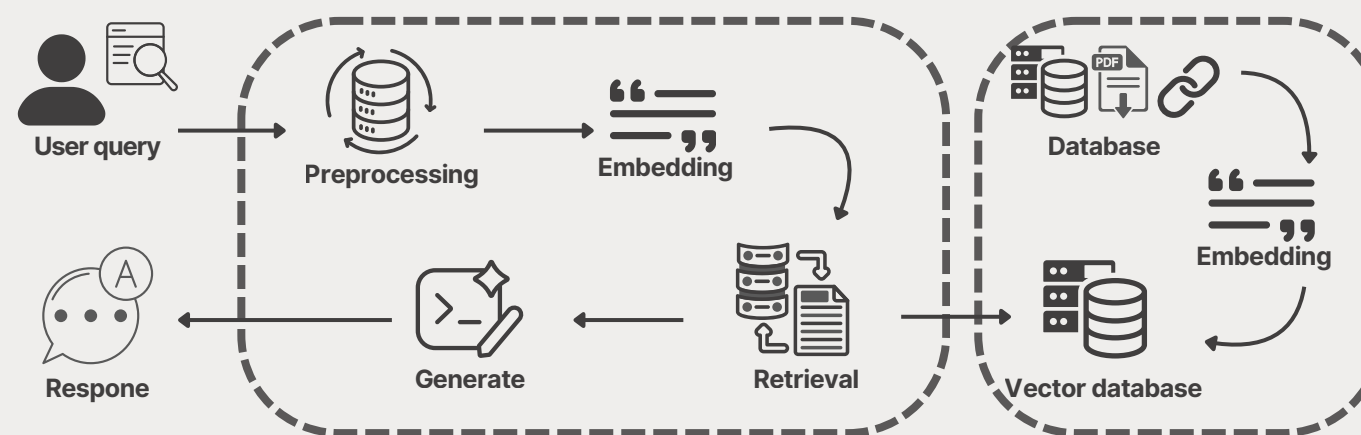
The classification identifies the range of NSWs in our work. It also provides possible type of NSWs in our training set. Each of the type has a specific way to convert the orthography into pronounceable form.

Subcategories of NUMBERS group are:

- *date:dmy* includes tokens which have full form of date. They contain 3 numbers which are separated by *dot* or *slash* operator. For example, 12/01/1989 is the same as 12.01.1989 and 12-01-1989. The first number represents a day in a month. The second number describes a month in a year
- *date:dy* for tokens having “day and month” construction



# Introduction



## Input:

- User-provided symptoms described in natural language

## RAG:

- **PhoBERT** + Article retrieval + **Vistral-7B-Chat**( generating)

## Output:

- Recommended medical department for examination
- General healthcare advice (non-diagnostic)

# Dataset

## BUILDING DATASET

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- Search for health articles and Q&A links from **trustworthy online sources**.
- Crawl **articles and Q&A** using **Selenium** and **BeautifulSoup**



- Preprocessing
- Article format: **'title', 'introduction', 'heading', 'content'**.

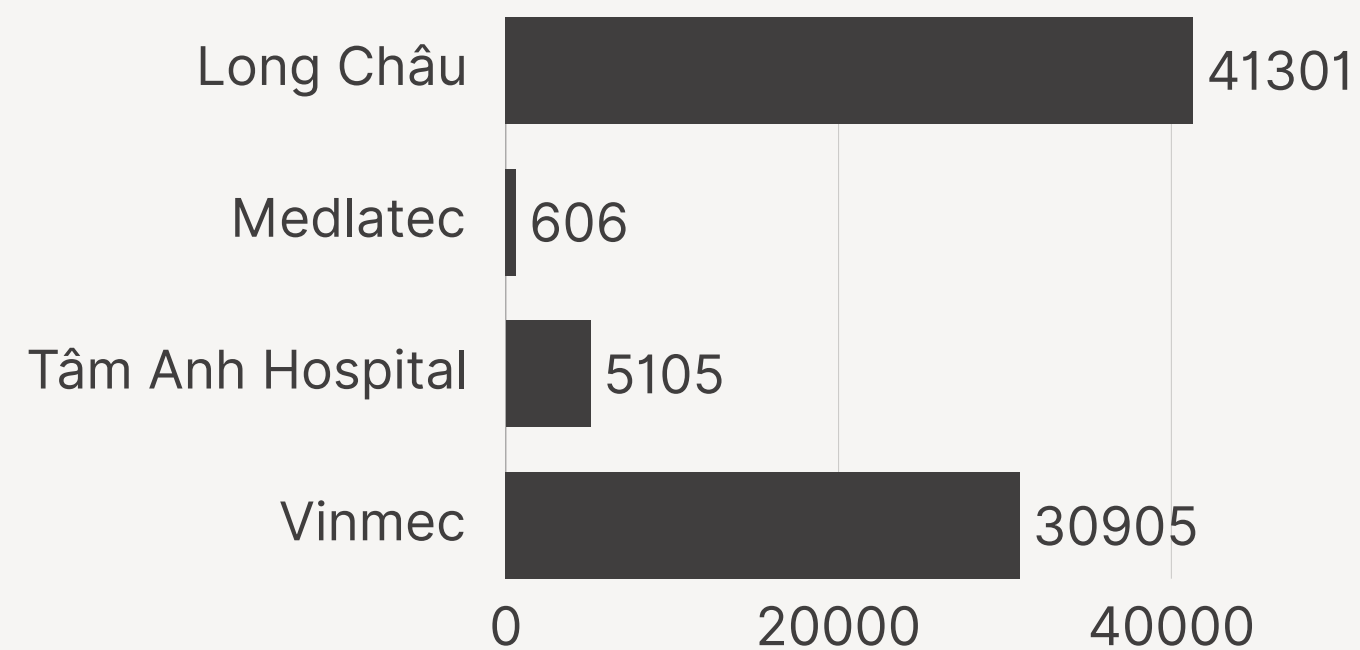


- **Topic labeling** for Q&A
- Add **'id'**

# Dataset

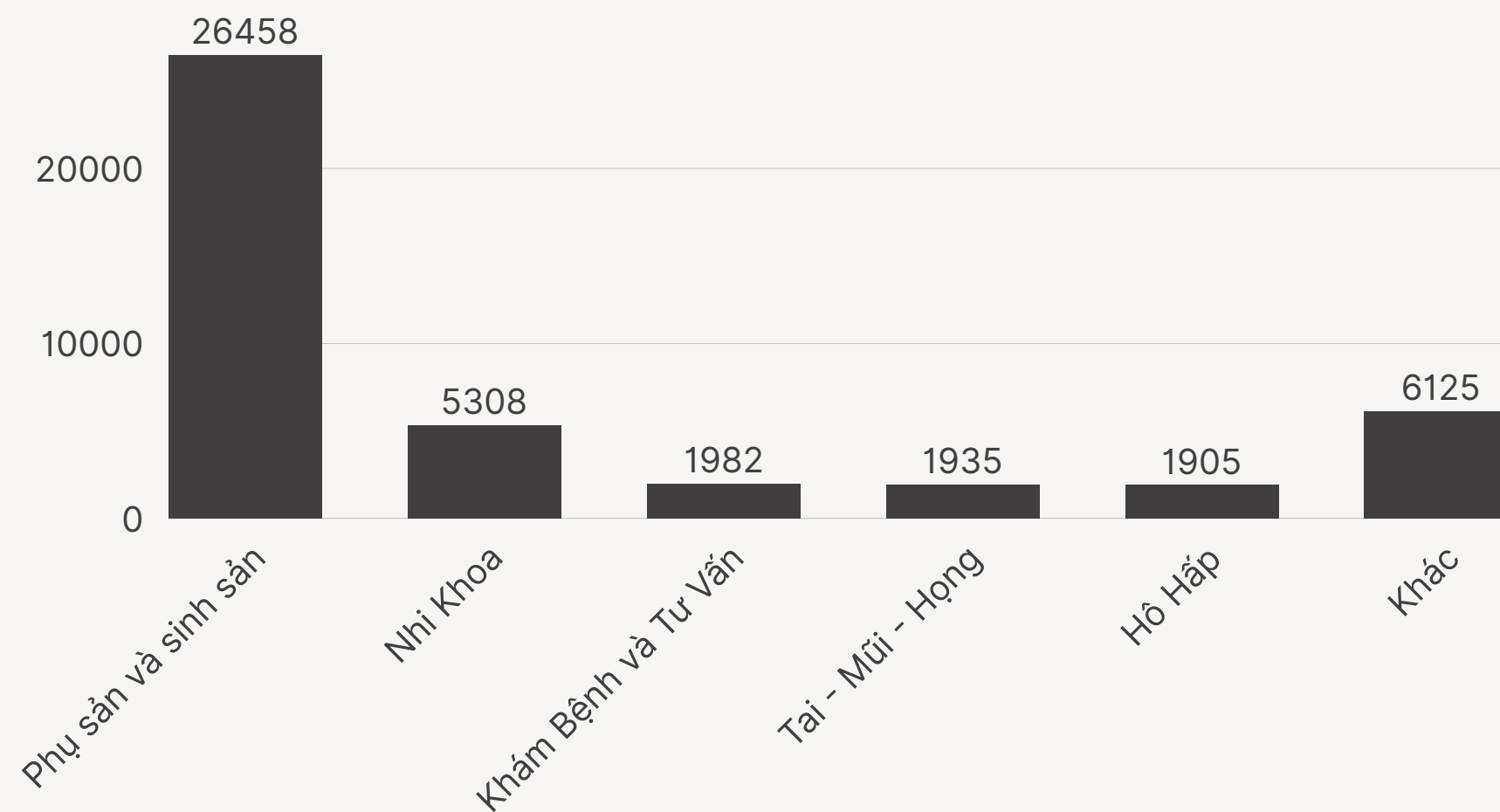
## ARTICLE DATASET

- Total: 77,917 articles



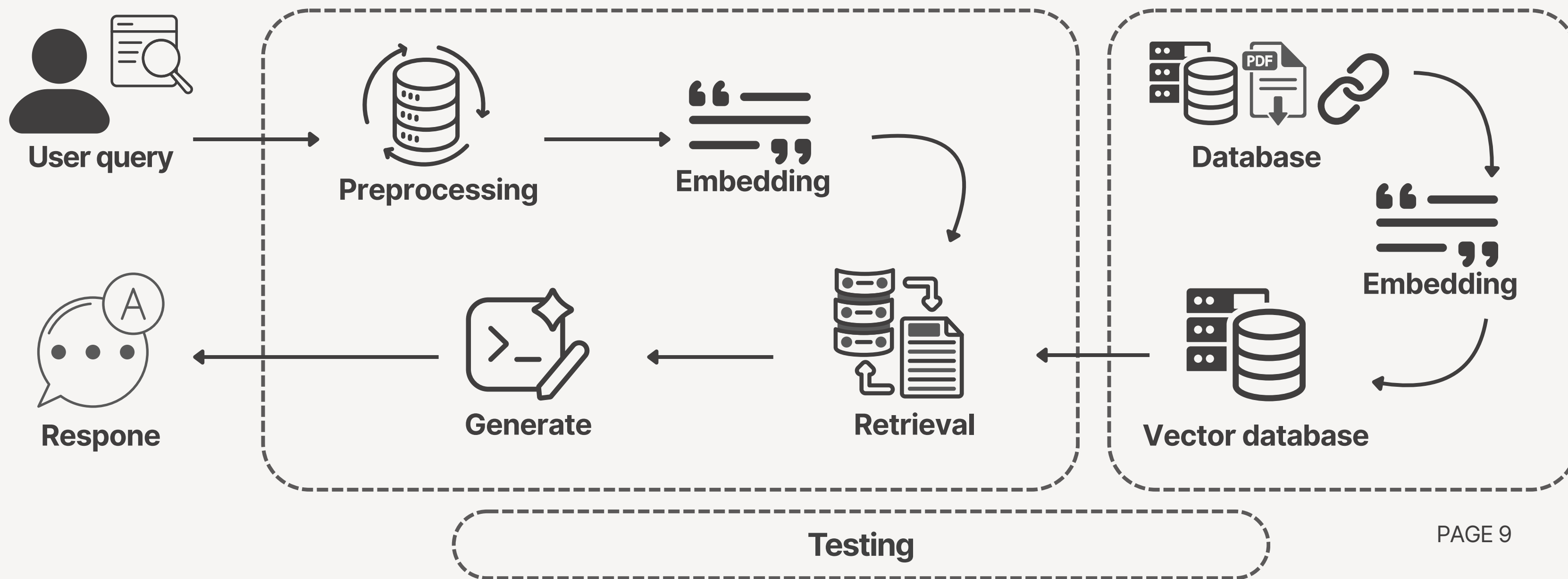
## Q&A DATASET

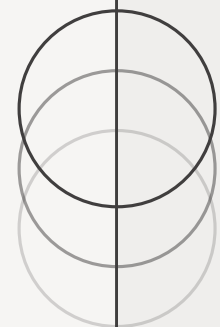
- Total: 60,234 Q&A





# Methodology





# Experiments

PREPROCESSING + EMBEDDING + INDEXING

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## Preprocessing:

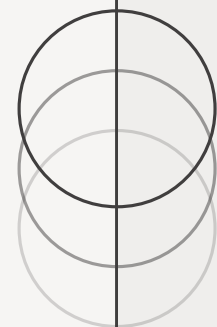
- **Tokenize and normalize** Vietnamese text using Underthesea

## Senmatic Embedding:

- Encode text with **PhoBERT** (vinai/phobert-base) as the retrieval encoder

## Indexing:

- Build **Approximate Nearest Neighbor** (ANN) indexes using **hnswlib** (HNSW) for Q/A and article datasets



# Experiments

## RETRIEVAL + ACTION EXTRACTION

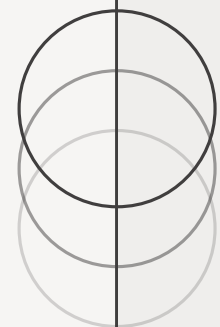
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### Retrieval:

- Convert user query into **embedding using PhoBERT**
- Retrieve **top-K relevant** Q/A pairs and medical articles from ANN indexes based on cosine similarity

### Action Extraction:

- Extract actionable **medical advice from Q/A answers** (sentences with explicit recommendations)
- Select the **most relevant article passages** to provide additional medical context



# Experiments

## RE-RANKING + GENERATION

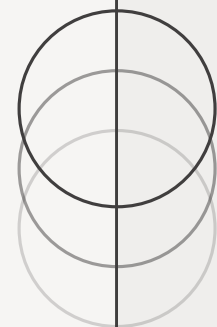
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### Re-ranking (Article Passage Selection):

- Re-rank retrieved candidates using: **Semantic similarity** (embedding-based) and **Lexical overlap**
- Apply **predefined thresholds and weights** to filter irrelevant results

### Generation:

- **Aggregate extracted Q/A** actions and article snippets
- Generate a concise and natural response using **Vistral-7B-Chat** with fallback models



# Experiments

## EVALUATION

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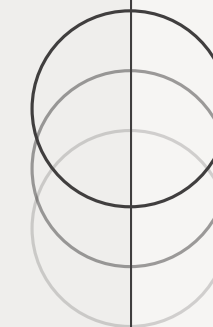
- Perform semantic evaluation using embedding-based similarity
- Metrics: **semantic\_recall@k**, **semantic\_mrr@k**, **action\_found**, **specialty\_acc**, **article\_found**.



# Results

## SEMANTIC EVALUATION

| Metric                           | Threshold   | k=5           | k=10          | Delta ( $\Delta$ ) |
|----------------------------------|-------------|---------------|---------------|--------------------|
| Semantic Recall,<br>Semantic MRR | [0; 0.50]   | 94.40%        | <b>96.80%</b> | 2.40%              |
|                                  | 0.65        | 87.20%        | <b>89.60%</b> | 2.40%              |
|                                  | 0.7         | <b>73.60%</b> | 72.80%        | -0.80%             |
|                                  | 0.75        | 47.20%        | <b>51.20%</b> | 4%                 |
|                                  | 0.8         | 14.40%        | <b>17.60%</b> | 3.20%              |
|                                  | 0.85        | <b>0.80%</b>  | <b>0.80%</b>  | 0%                 |
|                                  | [0.90;1.00] | 0.00%         | 0.00%         | 0%                 |
| Specialty<br>Accuracy            | [0.00;1.00] | <b>52.00%</b> | 45.60%        | -6.40%             |
| Action Found                     | [0.00;1.00] | 94.40%        | <b>96.80%</b> | 2.40%              |
| Article Found                    | [0.00;1.00] | 94.40%        | 94.40%        | 0%                 |



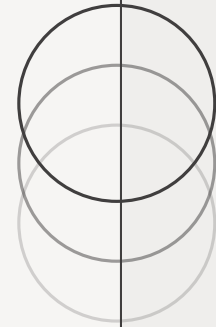
# Conclusion

## CURRENT LIMITATIONS

- **Topic imbalance**
- **Corpus** dependency
- **Context limitation**
- LLM **hallucination risk**
- **Static** thresholds & weights

## FUTURE WORK

- **Fine-tune** encoder
- Bi-encoder + Cross-encoder hybrid
- **Source citation, confidence scoring, and verification** of generated content
- Threshold tuning via validation set and grid search for alpha/beta/gamma
- **Medical content classifiers** and **harmful-content filtering**
- **Cache embeddings** and indexes to improve system efficiency



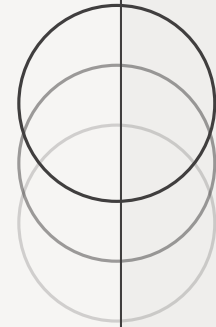
# THANK YOU

FOR LISTENING

Feel free to ask any questions!



DS300.Q11



**Khoa Khoa học  
và Kỹ thuật Thông tin**

# DEMO

GROUP 12

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